auto_cleaner

Capabilities & Proof of Work

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What it is

auto_cleaner ingests any messy tabular dataset (CSV, Parquet, JSON, SQL, FITS, netCDF), produces a mathematically clean, ML-ready dataset, and emits a comprehensive analysis report (interactive HTML, Markdown and PDF). It is built end-to-end on polars (no pandas in the engine) with DuckDB for SQL. Crucially, it inspects each dataset and AUTO-SPECIALISES: it detects the data's archetype and runs the analyses that actually fit.

Pipeline

ingest -> validate -> standardise -> impute -> outliers -> downcast -> profile -> auto-specialise -> inference / modelling / FDA / extended stats -> report.

Capability matrix

Ingestion: CSV/TSV/Parquet/JSON/NDJSON, DuckDB SQL, FITS (astronomy), netCDF (climate); auto format/delimiter/encoding/header detection; streaming/out-of-core.

Cleaning: safe dtype downcasting; imputation (median/mean/KNN/time-series ffill); outliers (IQR, Z-score, Isolation Forest); datetime/numeric-string/whitespace/categorical standardisation.

EDA: per-column profile, skewness, kurtosis, correlation & covariance, missingness, data-health warnings; HTML + Markdown + PDF reports.

Visualisation: interactive offline Plotly charts (histograms, frequency bars, scatter matrix, boxplots, correlation heatmap, missingness) + standalone PNG export.

Auto-specialisation: detects time-series, geospatial, text-heavy, high-dimensional/embeddings, survey, wide/omics, image-reference archetypes and routes modules accordingly.

Advanced: normality tests (Shapiro/D'Agostino/Jarque-Bera/Anderson-Darling), Box-Cox/Yeo-Johnson transforms, VIF + PCA, target-aware feature relevance (mutual information, ANOVA F).

Inference: bootstrap confidence intervals; auto-selected group tests (t/Welch/Mann-Whitney/ANOVA/Kruskal/chi2) WITH effect sizes (Cohen's d, Cliff's delta, eta^2); BH-corrected correlation significance; OLS/logit regression with p-values and CIs.

Modelling: cross-validated baselines (dummy/linear/forest/boosting) with metrics, permutation importance and a leakage check - a benchmark for a human, not a deployable model.

Extended statistics: robust means (geometric/harmonic/trimmed/winsorized/MAD/Huber); rank, partial & categorical associations (Spearman/Kendall/partial/Cramer's V/eta); distribution fitting (AIC/BIC); multivariate (Mahalanobis, clustering, MANOVA, UMAP); classical NLP (LDA topics, sentiment); Bayes factors; survival (KM, Cox); survey (Cronbach, weights).

Functional (FDA): smoothing + functional PCA of time-indexed curves.

Hardening: typed validation + schema enforcement, expanded test suite, GitHub Actions CI.

Honest scope & limitations

- Modelling and inference are baselines/diagnostics for a human to validate, not final causal claims or production models.
- Domain-aware via heuristics, not a domain expert (no automatic survey weighting design, omics normalisation, astrometric calibration).
- Tabular focus; no computer vision / audio / graph deep learning. Neural text embeddings are opt-in.
- A clean, tested package - not a library hardened by years of production edge-cases.

Proof of work

Every number below comes from a real auto_cleaner run on the public auto-mpg ('cars') dataset (406 rows x 9 columns), not from hand-picked examples.

Run summary

Rows in -> out: 406 -> 406
Memory in -> out: 31.1 KB -> 17.3 KB (-44.4%)
Data-health warnings raised: 3 Elapsed: 18.76s

Auto-specialisation

Detected: generic tabular (60%)
Auto-modules run: inference, model

Data-health warnings (verbatim)

- Feature 'Name' is high-cardinality (311 levels) ? encoding may explode dimensionality
- High collinearity between 'Cylinders' and 'Displacement' (r=+0.95)
- High collinearity between 'Displacement' and 'Weight_in_lbs' (r=+0.93)

Baseline models (5-fold cross-validation)

- Dummy (mean): R2=-0.9803, RMSE=8.275, MAE=7.026
- Linear: R2=0.3155, RMSE=4.754, MAE=3.84
- Random Forest: R2=0.4093, RMSE=4.417, MAE=3.447 <- best
- Gradient Boosting: R2=0.3989, RMSE=4.498, MAE=3.531

Significant correlations (Benjamini-Hochberg corrected)

- Cylinders ~ Displacement: r=+0.952, p(adj)=2.9e-208
- Displacement ~ Weight_in_lbs: r=+0.932, p(adj)=7.096e-180
- Displacement ~ Horsepower: r=+0.897, p(adj)=2.217e-144
- Cylinders ~ Weight_in_lbs: r=+0.895, p(adj)=2.532e-143
- Horsepower ~ Weight_in_lbs: r=+0.864, p(adj)=3.332e-122

OLS regression on 'Miles_per_Gallon' (R2=0.6882, n=406)

- const: coef=46.81, p=2.748e-51, 95% CI [41.55, 52.07]
- Weight_in_lbs: coef=-0.005386, p=1.436e-10, 95% CI [-0.006994, -0.003778]
- Horsepower: coef=-0.04056, p=0.01445, 95% CI [-0.07302, -0.0081]
- Cylinders: coef=-0.3396, p=0.4136, 95% CI [-1.155, 0.4762]
- Acceleration: coef=-0.07016, p=0.5727, 95% CI [-0.3145, 0.1741]
- Displacement: coef=-0.0002426, p=0.979, 95% CI [-0.01837, 0.01789]

Best-fit distributions (lowest AIC)

- Miles_per_Gallon: weibull_min (AIC 2,776, KS p 0.4357)
- Displacement: weibull_min (AIC 4,739, KS p 0.0001)
- Horsepower: lognorm (AIC 4,007, KS p 0.0336)
- Weight_in_lbs: weibull_min (AIC 6,555, KS p 0.0671)
- Acceleration: lognorm (AIC 1,991, KS p 0.5804)

Multivariate

Mahalanobis outliers: 23; clusters: 2 (silhouette 0.546); 2-D PCA variance: 0.916

Alongside this document, each run also produces an interactive HTML report, a Markdown report, a per-dataset PDF, and standalone PNG charts.